

Course 47211 “Electrochemical Energy Storage and Power2X”

Project: Design the future Danish 100% Sustainable Energy System

Assess different scenarios for transitioning to a 100% Sustainable Energy System in DK relying on biomass and (intermittent) renewable energy from solar/wind, by exploiting appropriate electrochemical energy storage/conversion and Power2X devices.

- Final report (max 5 pages not including references) due <3 days after lecture 13, i.e. before December 6th, 12h. The report should contain the following sections (free subsection structure):
 - Abstract (max 1000 characters (with spaces))
 - Problem statement and approach (max 1000 characters (with spaces))
 - Results & Discussion
 - Conclusions & Outlook
 - References

The reports are checked for plagiarism
- Selected oral presentations (from volunteering and/or randomly picked groups) on lecture 13, December 3rd.

AI use is allowed for searching, ideation, calculations, structure, drafting, editing, etc. but each one of the group members is expected to i) know how the report calculations were made, ii) own all decisions and the final text, iii) verify AI output and cite reliable sources. The report should contain a statement/acknowledgement on how, to what extent, and which AI tool was used.

Project guidelines

- 1) Use the following values (PJ) concerning the total annual energy consumption in DK, and its distribution between households, heavy transport, light transport and industry.

Total	Households*	Industry**	Light Transport	Heavy Transport		
				Trucks	Shipping***	Aviation
730	300	150+30	100	50	50	50

* including commercial and public services, ** including agriculture and 30PJ of data centers expected by 2030, ***not including 500 PJ of international shipping under Danish flag
(Source: <https://ens.dk/analyser-og-statistik/aarlig-energistatistik> and others)
Use a value of 150 PJ for the available biomass (and sustainable trash) in DK.

- 2) Collect information about the total annual wind (onshore + offshore) and solar renewable electricity production and the total electricity consumption in DK, by integrating the hourly generation and consumption time series over the course of last year (01.01.2025 – 31.12.2025).
<https://www.energidataservice.dk/tso-electricity/productionconsumptionsettlement>
For the consumption you should use the gross consumption data that accounts for all the electricity import and export figures and losses in the electricity grid. In all cases, you need to add the DK1 and DK2 contributions.
- 3) Scale the values of wind and solar generation time-series (retaining their ratio) to match 100% of the cumulative annual energy consumption in DK (given in the Table of step1) minus the biomass contribution (given in step1). Scale the electricity consumption time-series values to also match the cumulative direct electricity consumption (assumed to be the sum of households & industry). Plot electricity production (wind + solar), direct electricity consumption (households & industry), and prosumption (production - consumption) time series. Provide the deduced scaling factors in the Caption of this Figure 1.

- 4) Set up different conceptual scenarios for balancing resources (electricity and biomass) with demand (electricity and fuel), making use of the electrochemical energy storage and P2X conversion technologies introduced in the course (or that you have identified from additional sources in the literature). Consider DK to be isolated (i.e. no exchange with neighboring regions). Take for granted that heavy transport (other than trucks) cannot be electrified. Your scenario needs to include at least one electrochemical energy conversion and at least one P2X technology. Revise your direct electricity consumption time series to match your scenario(s) if needed.
- 5) Analyze the time-series to derive storage/conversion requirements over the course of the year. What is the max energy storage capacity required over the course of the year? What is the max power capacity required to balance supply with demand?
Here you need to consider:
 - whether parts of the transport sector can be electrified and what time profile this electricity consumption will have
 - the operational profile strategy (and associated electricity consumption) of the P2X components in the system
 - how will biomass be used?
 - can P2X and biomass be exploited to minimize electricity storage and recovery requirements?
 Plot time series of presumption and foreseen operating profiles (time series) of P2X and biomass utilization. Provide numbers of storage capacity requirements (energy and power in SI units) in the Caption of this Figure 2.
- 6) Estimate required scale/capacity for the selected technologies, and calculate the overall energy efficiency of your solution and the levelized cost of energy achieved, based on performance and economy figures for the selected technologies (listed in an Addendum; not included in the 5 page limit). Feel free to make use of the corresponding technoeconomic descriptions you have accumulated from previous assignments in order to optimize your scenario in terms of the overall levelized cost of energy. Assume a flat electricity price of 0.03 €/kWh.
- 7) Compare pros and cons of the different scenarios you have assessed.
 - Present comparisons using Sankey diagrams and key performance indicators (LCOE, efficiency)
 - How does your fuel cost compare to MeOH imported from Brazil (see battle 2)

Additional considerations (optional)

- Factor-in the additional electricity needs to compensate for efficiency losses in your scenario(s).
- Could you imagine a system that would enable excess e-fuel production for export?
- Use dynamic efficiency of the electrochemical conversion devices (taking into account the actual operating point at each instance rather than a nominal fixed efficiency value).
- Factor-in cost of BEVs and/or FCEVs and charging/fueling infrastructure if relevant in your scenario.
- Compare the isolated scenario to a scenario where exchange of electricity with neighboring countries can offer a means of balancing production with consumption (take into consideration grid losses and associated infrastructure CAPEX).
- Compare the isolated scenario to a scenario where biomass import is allowed.
- A large part of the energy consumption for households and Industry is heat. Could some of this be covered by efficiency losses in your system?